

HAILUOTO KESKIKYLA, Finland

WMO: 028740

Lat: 65.020N

Lon: 24.727E

Elev: 6

StdP: 101.25

Time Zone: 2.00 (EUE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-23.9	-29.9	0.2	-27.0	-26.4	0.3	-23.9	9.4	-0.4	8.5	-0.4	0.7	110	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.0	18.4	22.9	17.4	21.2	16.5	19.7	23.5	18.4	21.7	17.3	20.3	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.2	21.4	17.1	12.2	20.0	16.0	11.3	19.0	56.1	23.8	52.3	21.7	48.7	20.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.4	-28.6	26.8	4.2	2.9	-31.6	28.9	-34.0	30.6	-36.3	32.3	-39.3	34.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-28.7	21.0	4.1	1.8	-31.7	22.3	-34.1	23.3	-36.4	24.3	-39.4	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-8.7	-8.5	-4.7	1.0	7.6	12.0	15.8	14.1	9.5	3.3	-0.7	-4.1
	DBStd	9.60	7.31	7.45	5.33	3.29	4.13	3.37	3.15	3.20	3.25	3.89	4.59	6.35
	HDD10.0	2956	578	518	454	271	99	16	1	6	47	208	321	437
	HDD18.3	5572	837	751	713	521	334	192	92	137	263	466	571	696
	CDD10.0	446	0	0	0	0	24	77	180	131	33	1	0	0
	CDD18.3	20	0	0	0	0	1	2	13	4	0	0	0	0
	CDH23.3	157	0	0	0	0	7	26	99	26	0	0	0	0
	CDH26.7	25	0	0	0	0	0	5	16	4	0	0	0	0
Wind	WSAvg	2.9	2.8	3.0	3.2	2.9	3.2	3.0	2.5	2.6	2.8	3.0	3.1	3.1
Precipitation	PrecAvg	520	38	33	30	28	38	42	63	55	49	50	49	43
	PrecMax	694	73	69	81	78	84	102	119	117	90	97	84	86
	PrecMin	371	14	14	5	8	6	2	3	4	11	7	10	12
	PrecStd	85	17	17	17	16	18	27	30	26	24	21	19	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	4.0	5.5	14.0	24.1	26.9	28.7	26.2	19.6	12.8	8.8	5.1
		MCWB	2.2	2.5	3.0	7.7	15.0	17.7	19.8	18.9	15.0	10.8	7.6	4.2
	2%	DB	2.0	2.8	3.8	10.6	20.8	22.5	25.9	23.1	17.1	11.0	7.6	4.2
		MCWB	1.4	1.7	1.8	5.7	14.0	16.0	19.2	17.8	14.0	9.6	6.7	3.4
	5%	DB	1.1	1.6	2.8	7.9	17.3	20.2	23.8	21.3	15.7	9.8	6.2	3.4
		MCWB	0.7	0.8	1.3	4.4	11.3	14.5	18.1	16.9	13.3	8.7	5.5	2.6
	10%	DB	0.3	0.5	1.8	5.9	14.7	18.3	22.1	19.7	14.5	8.9	4.9	2.5
		MCWB	0.0	0.0	0.6	3.2	9.9	13.6	17.4	16.4	12.7	7.8	4.3	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.9	3.4	8.7	16.1	19.4	21.7	20.6	16.0	11.4	8.0	4.6
		MCDB	2.8	3.8	5.1	12.0	22.4	25.2	25.5	24.4	18.2	12.4	8.5	4.8
	2%	WB	1.6	1.8	2.2	6.1	14.1	17.2	20.3	18.9	14.7	9.9	6.7	3.6
		MCDB	1.9	2.6	3.3	9.5	19.7	21.0	24.3	21.6	16.4	10.8	7.4	4.0
	5%	WB	0.7	0.9	1.5	4.7	12.1	15.6	19.1	17.8	13.7	9.0	5.6	2.7
		MCDB	1.0	1.5	2.4	7.7	16.5	18.8	22.6	20.2	15.1	9.7	6.1	3.3
	10%	WB	0.0	0.1	0.8	3.5	10.5	14.3	18.0	16.7	12.9	7.9	4.4	2.0
		MCDB	0.4	0.5	1.7	5.6	13.7	17.5	21.4	19.0	14.3	8.8	4.9	2.4
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.9	7.1	7.5	8.6	8.3	8.6	8.2	6.7	5.1	4.0	5.1
		MCDBR	4.5	4.6	5.8	11.5	13.8	12.0	12.0	11.2	8.5	5.5	4.1	3.9
		MCWBR	4.5	4.1	4.4	7.4	7.4	6.4	6.1	6.2	5.8	4.6	3.9	3.5
	5% WB	MCDBR	4.6	4.4	4.9	10.2	12.8	9.8	10.3	9.1	7.4	5.1	4.0	3.8
		MCWBR	4.6	4.0	3.9	6.9	7.5	5.9	5.8	5.9	5.5	4.5	3.9	3.4
Clear-Sky Solar Irradiance	taub	0.209	0.276	0.305	0.343	0.347	0.350	0.370	0.354	0.331	0.308	0.344	0.354	
	taud	1.951	2.097	2.141	2.190	2.410	2.483	2.450	2.486	2.526	2.353	2.545	2.590	
	Ebn at Noon	293	596	758	819	861	866	833	813	755	577	269	112	
	Edh at Noon	23	60	91	111	99	94	95	83	65	48	21	12	
All-Sky Solar Radiation	RadAvg	0.11	0.62	2.09	3.79	4.96	5.56	5.13	3.80	2.08	0.76	0.16	0.03	
	RadStd	0.01	0.12	0.21	0.30	0.47	0.44	0.46	0.36	0.21	0.12	0.03	0.00	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page